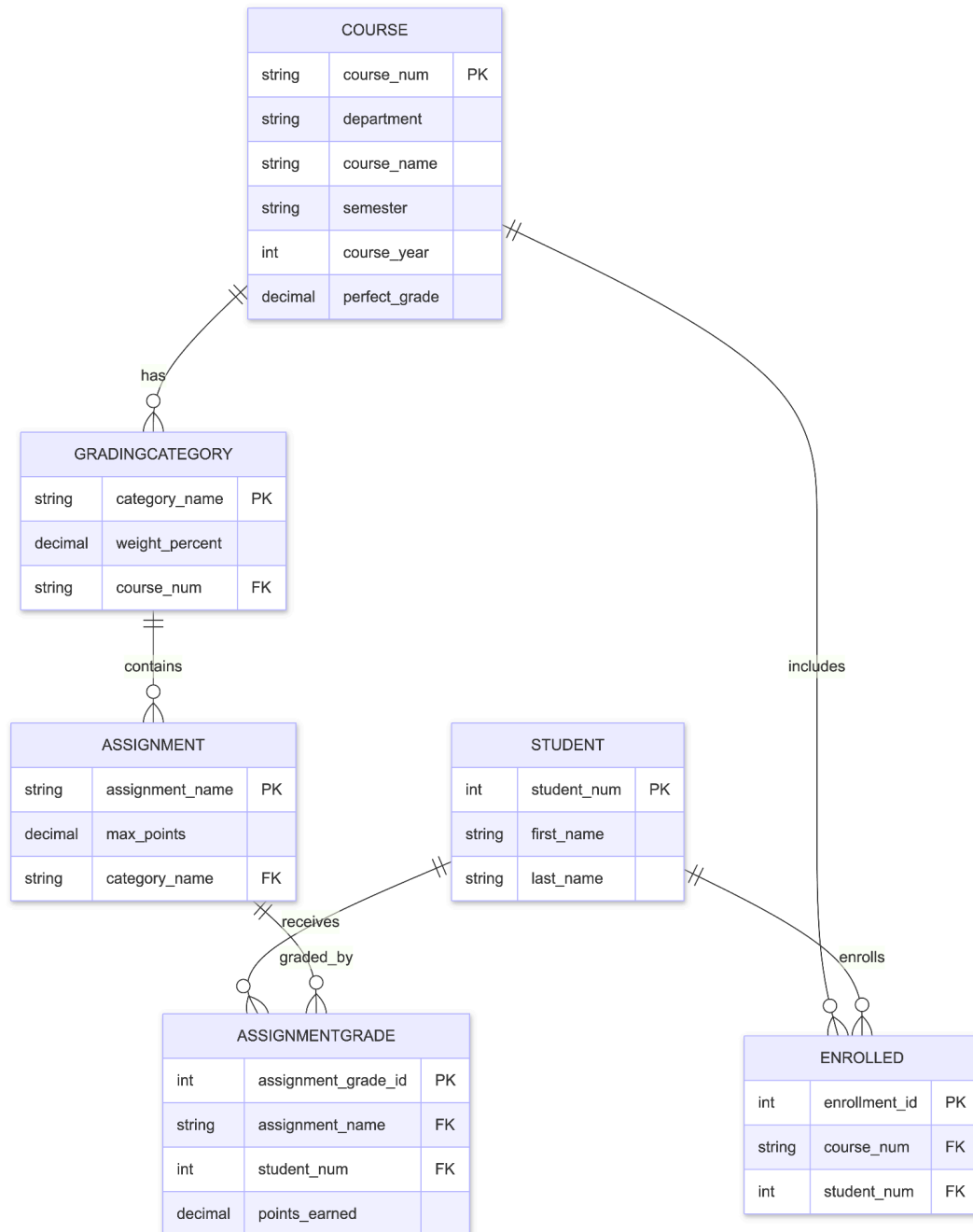


Project Submissions Document (1-4, 7)

ER Diagram



Task 2 Commands:

```
DROP DATABASE IF EXISTS gradebook_db;
CREATE DATABASE gradebook_db;
USE gradebook_db;

-- TABLES

CREATE TABLE Course (
    course_id INT PRIMARY KEY AUTO_INCREMENT,
    department VARCHAR(10) NOT NULL,
    course_num VARCHAR(10) NOT NULL,
    course_name VARCHAR(100) NOT NULL,
    semester VARCHAR(20) NOT NULL,
    course_year INT NOT NULL,
    perfect_grade DECIMAL(5,2) NOT NULL DEFAULT 100.00
);

CREATE TABLE GradingCategory (
    grading_category_id INT PRIMARY KEY AUTO_INCREMENT,
    course_id INT NOT NULL,
    category_name VARCHAR(40) NOT NULL,
    weight_percent DECIMAL(5,2) NOT NULL
);

CREATE TABLE Assignment (
    assignment_id INT PRIMARY KEY AUTO_INCREMENT,
    grading_category_id INT NOT NULL,
    assignment_name VARCHAR(100) NOT NULL,
    max_points DECIMAL(6,2) NOT NULL
);

CREATE TABLE Student (
    student_num INT PRIMARY KEY AUTO_INCREMENT,
    first_name VARCHAR(40),
    last_name VARCHAR(40)
);

CREATE TABLE Enrollment (
    enrollment_id INT PRIMARY KEY AUTO_INCREMENT,
    course_id INT,
    student_num INT
);
```

```
CREATE TABLE AssignmentGrade (  
    assignment_grade_id INT PRIMARY KEY AUTO_INCREMENT,  
    assignment_id INT,  
    student_num INT,  
    points_earned DECIMAL(6,2)  
);
```

```
-- INSERT DATA
```

```
INSERT INTO Course VALUES (1, 'CS', '340', 'Database  
Systems', 'Fall', 2026, 100);
```

```
INSERT INTO GradingCategory VALUES  
(1, 1, 'Participation', 10),  
(2, 1, 'Homework', 20),  
(3, 1, 'Tests', 50),  
(4, 1, 'Projects', 20);
```

```
INSERT INTO Assignment VALUES  
(1, 1, 'Participation Log', 100),  
(2, 2, 'HW1', 100),  
(3, 2, 'HW2', 100),  
(4, 3, 'Midterm', 100),  
(5, 3, 'Final', 100),  
(6, 4, 'Course Project', 100);
```

```
INSERT INTO Student VALUES  
(1, 'Alice', 'Anderson'),  
(2, 'Robert', 'Quincy'),  
(3, 'Carol', 'Smith');
```

```
INSERT INTO Enrollment VALUES  
(1, 1, 1), (2, 1, 2), (3, 1, 3);
```

```
INSERT INTO AssignmentGrade VALUES  
(1, 1, 1, 90), (2, 2, 1, 88), (3, 3, 1, 92),  
(4, 4, 1, 85), (5, 5, 1, 91), (6, 6, 1, 89);
```

Task 3 Output:

```

Course
course_id  department  course_num  course_name  semester  course_year  perfect_grade
1          CS      340        Database Systems  Fall    2026      100.00

tbl
GradingCategory
grading_category_id  course_id  category_name  weight_percent
1          1      Participation  10.00
2          1      Homework      20.00
3          1      Tests         50.00
4          1      Projects       20.00

tbl
Assignment
assignment_id  grading_category_id  assignment_name  max_points
1          1      Participation Log  100.00
2          2      HW1              100.00
3          2      HW2              100.00
4          3      Midterm          100.00
5          3      Final            100.00
6          4      Course Project   100.00

tbl
Student
student_num  first_name  last_name  email
1          Alice  Anderson  alice.anderson@example.edu
2          Robert  Quincy  robert.quincy@example.edu
3          Carol  Smith  carol.smith@example.edu

tbl
Enrollment
enrollment_id  course_id  student_num
1          1      1
2          1      2
3          1      3

tbl
AssignmentGrade
assignment_grade_id  assignment_id  student_num  points_earned
1          1      1      90.00
2          2      1      88.00
3          3      1      92.00
4          4      1      85.00
5          5      1      91.00
6          6      1      89.00
7          1      2      95.00
8          2      2      70.00
9          3      2      75.00
10         4      2      80.00
11         5      2      82.00
12         6      2      78.00
13         1      3      80.00
14         2      3      90.00
15         3      3      95.00
16         4      3      60.00
17         5      3      88.00
18         6      3      92.00

department  course_num  semester  course_year  category_name  assignment_name  average_score  highest_score  lowest_score
CS          340      Fall    2026      Homework      HW1      84.67      92.00      72.00

student_num  first_name  last_name  email
1          Alice  Anderson  alice.anderson@example.edu
2          Robert  Quincy  robert.quincy@example.edu
3          Carol  Smith  carol.smith@example.edu

student_num  first_name  last_name  category_name  assignment_name  max_points  points_earned
1          Alice  Anderson  Homework      HW1      100.00      90.00
1          Alice  Anderson  Homework      HW2      100.00      92.00
1          Alice  Anderson  Homework      HW3      100.00      NULL
1          Alice  Anderson  Participation  Participation Log  100.00      90.00
1          Alice  Anderson  Projects      Course Project  100.00      89.00
1          Alice  Anderson  Tests         Final          100.00      91.00
1          Alice  Anderson  Tests         Midterm        100.00      85.00
2          Robert  Quincy  Homework      HW1      100.00      72.00
2          Robert  Quincy  Homework      HW2      100.00      77.00
2          Robert  Quincy  Homework      HW3      100.00      NULL
2          Robert  Quincy  Participation  Participation Log  100.00      95.00
2          Robert  Quincy  Projects      Course Project  100.00      78.00
2          Robert  Quincy  Tests         Final          100.00      82.00
2          Robert  Quincy  Tests         Midterm        100.00      80.00
3          Carol  Smith  Homework      HW1      100.00      92.00
3          Carol  Smith  Homework      HW2      100.00      95.00
3          Carol  Smith  Homework      HW3      100.00      NULL
3          Carol  Smith  Participation  Participation Log  100.00      80.00
3          Carol  Smith  Projects      Course Project  100.00      92.00
3          Carol  Smith  Tests         Final          100.00      88.00
3          Carol  Smith  Tests         Midterm        100.00      60.00

student_num  first_name  last_name  final_grade
1          Alice  Anderson  88.94
2          Robert  Quincy  80.63
3          Carol  Smith  81.71

student_num  first_name  last_name  final_grade_drop_lowest
1          Alice  Anderson  90.68
2          Robert  Quincy  81.60
3          Carol  Smith  89.26

```

Task 4 Commands:

```
SELECT
    AVG(points_earned) AS average,
    MAX(points_earned) AS highest,
    MIN(points_earned) AS lowest
FROM AssignmentGrade
WHERE assignment_id = 2;
```

Task 5 Commands:

```
SELECT s.student_num, s.first_name, s.last_name
FROM Enrollment e
JOIN Student s ON e.student_num = s.student_num
WHERE e.course_id = 1;
```

Task 6 Commands:

```
SELECT s.first_name, s.last_name, a.assignment_name, ag.points_earned
FROM Student s
JOIN AssignmentGrade ag ON s.student_num = ag.student_num
JOIN Assignment a ON ag.assignment_id = a.assignment_id;
```

Task 7 Commands:

```
INSERT INTO Assignment (grading_category_id, assignment_name, max_points)
VALUES (2, 'HW3', 100);
```

Task 8 Commands:

```
UPDATE GradingCategory
SET weight_percent = CASE
    WHEN category_name = 'Homework' THEN 18
    WHEN category_name = 'Tests' THEN 52
    ELSE weight_percent
END;
```

Task 9 Commands:

```
UPDATE AssignmentGrade
SET points_earned = points_earned + 2
WHERE assignment_id = 2;
```

Task 10 Commands:

```
UPDATE AssignmentGrade ag
JOIN Student s ON ag.student_num = s.student_num
SET ag.points_earned = ag.points_earned + 2
WHERE s.last_name LIKE '%Q%';
```

Task 11 Commands:

```
SELECT s.student_num, s.first_name, s.last_name,
SUM(ag.points_earned) AS total_score
FROM Student s
JOIN AssignmentGrade ag ON s.student_num = ag.student_num
GROUP BY s.student_num;
```

Task 12 Commands:

```
WITH base AS (
  SELECT
    ag.student_num,
    gc.grading_category_id,
    gc.category_name,
    gc.weight_percent,
    a.assignment_id,
    ag.points_earned,
    a.max_points,
    COUNT(*) OVER (
      PARTITION BY ag.student_num, gc.grading_category_id
    ) AS assignments_in_category,
    ROW_NUMBER() OVER (
      PARTITION BY ag.student_num, gc.grading_category_id
```

```
        ORDER BY ag.points_earned ASC, a.assignment_id ASC
    ) AS rn_lowest_first
FROM AssignmentGrade ag
JOIN Assignment a ON a.assignment_id = ag.assignment_id
JOIN GradingCategory gc ON gc.grading_category_id =
a.grading_category_id
JOIN Course c ON c.course_id = gc.course_id
WHERE c.department = 'CS'
),
filtered AS (
    SELECT *
    FROM base
    WHERE assignments_in_category = 1
        OR rn_lowest_first > 1
),
per_category AS (
    SELECT
        student_num,
        grading_category_id,
        category_name,
        weight_percent,
        SUM(points_earned) AS earned,
        SUM(max_points) AS possible
    FROM filtered
    GROUP BY student_num, grading_category_id
)
SELECT
    s.student_num,
    s.first_name,
    s.last_name,
    ROUND(SUM((earned / possible) * weight_percent), 2) AS
final_grade_drop_lowest
FROM per_category pc
JOIN Student s ON s.student_num = pc.student_num
GROUP BY s.student_num
ORDER BY s.last_name;
```

Test Cases

Test Case 1: Verify Table Data Insertion

Description:

Check that all tables were created and populated correctly.

Query:

```
SELECT * FROM Student;
```

Expected Result:

3 students (Alice, Robert Quincy, Carol)

Actual Result:

Student				
student_num	first_name	last_name	email	
1	Alice	Anderson	alice.anderson@example.edu	
2	Robert	Quincy	robert.quincy@example.edu	
3	Carol	Smith	carol.smith@example.edu	
...				

Test Case 2: Average, Highest, Lowest Score

Description:

Verify correct calculation of statistics for an assignment.

Query:

```
SELECT AVG(points_earned), MAX(points_earned), MIN(points_earned)
FROM AssignmentGrade
WHERE assignment_id = 2;
```

Expected Result:

Correct average, highest, and lowest values

Actual Result:

department	course_num	semester	course_year	category_name	assignment_name	average_score	highest_score	lowest_score
CS	340	Fall	2026	Homework	Hw1	84.67	92.00	72.00

Test Case 3: Add +2 Points to All Students

Description:

Ensure all students' scores increase by 2 points.

Query:

```
UPDATE AssignmentGrade
SET points_earned = points_earned + 2
WHERE assignment_id = 2;
```

Expected Result:

All HW1 scores increase by 2

Actual Result:

student_num	first_name	last_name	category_name	assignment_name	max_points	points_earned
1	Alice	Anderson	Homework	HW1	100.00	90.00
1	Alice	Anderson	Homework	HW2	100.00	92.00
1	Alice	Anderson	Homework	HW3	100.00	NULL
2	Robert	Quincy	Homework	HW1	100.00	72.00
2	Robert	Quincy	Homework	HW2	100.00	77.00
2	Robert	Quincy	Homework	HW3	100.00	NULL
3	Carol	Smith	Homework	HW1	100.00	92.00
3	Carol	Smith	Homework	HW2	100.00	95.00
3	Carol	Smith	Homework	HW3	100.00	NULL

Test Case 4: Add +2 Points to Students with 'Q'

Description:

Only students with "Q" in last name should be updated.

Query:

```
UPDATE AssignmentGrade ag
JOIN Student s ON ag.student_num = s.student_num
SET ag.points_earned = ag.points_earned + 2
WHERE s.last_name LIKE '%Q%';
```

Expected Result:

Only Robert Quincy is affected

Actual Result:

2	Robert	Quincy	Homework	HW1	100.00	72.00		
2	Robert	Quincy	Homework	HW2	100.00	77.00		
2	Robert	Quincy	Homework	HW3	100.00	NULL		
2	Robert	Quincy	Participation	Participation Log			100.00	95.00
2	Robert	Quincy	Projects	Course Project	100.00	78.00		
2	Robert	Quincy	Tests	Final	100.00	82.00		
2	Robert	Quincy	Tests	Midterm	100.00	80.00		

Test Case 5: Final Grade Calculation

Description:

Verify weighted grade calculation.

Query:

```

SELECT
    s.student_num,
    s.first_name,
    s.last_name,
    ROUND(SUM((earned / possible) * gc.weight_percent), 2) AS final_grade
FROM Student s
JOIN Enrollment e ON s.student_num = e.student_num
JOIN Course c ON e.course_id = c.course_id
JOIN GradingCategory gc ON gc.course_id = c.course_id
JOIN (
    SELECT
        ag.student_num,
        a.grading_category_id,
        SUM(ag.points_earned) AS earned,
        SUM(a.max_points) AS possible
    FROM AssignmentGrade ag
    JOIN Assignment a ON ag.assignment_id = a.assignment_id
    GROUP BY ag.student_num, a.grading_category_id
) sub ON sub.student_num = s.student_num
    AND sub.grading_category_id = gc.grading_category_id
WHERE c.department = 'CS'
GROUP BY s.student_num, s.first_name, s.last_name;

```

Expected Result:

Each student has a final grade

Actual Result:

student_num	first_name	last_name	final_grade
1	Alice	Anderson	88.94
2	Robert	Quincy	80.63
3	Carol	Smith	81.71

Test Case 6: Drop Lowest Score

Description:

Ensure lowest assignment in each category is removed.

Expected Result:

Final grade slightly increases or changes correctly

Actual Result:

student_num	first_name	last_name	final_grade_drop_lowest
1	Alice	Anderson	90.68
2	Robert	Quincy	81.60
3	Carol	Smith	89.26